



CSCE 633-700: Machine Learning

Linear regression

Overview

Linear Regression

- Example
- Representation, Evaluation
- Optimization: Closed form solution via Ordinary Least Squares
- Optimization: Numerical solution via Gradient Descent
 - General gradient descent
 - Gradient descent for linear regression (batch, stochastic, mini-batch)
- Non-linear basis function for regression & Overfitting

[Parts of these slides have been adapted from K. Murphy (Machine Learning: A probabilistic perspective), Dr. Andrew Ng's Machine Learning course at Coursera, and CSCI567 Machine Learning (USC, Drs. Sha & Liu)]

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Linear Regression: Example



Apartment Amenities

Pet Policy

Dogs Allowed: Breed Restrictions May Apply.

- \$250 Fee
- 2 Pet Limit

Cats Allowed

- \$250 Fee
- 2 Pet Limit

Fitness & Recreation

- Fitness Center
- Pool
- Volleyball Court

Living Space

- Carpet
- Attic
- Crown Molding
- Views
- Walk-In Closets

Outdoor Space

- Balcony
- Patio
- Porch
- Yard
- Barbecue Area
- Barbecue/Grill

Parking

Surface Lot
6 spaces; Assigned Parking.

Services

- Maintenance on site
- Property Manager on Site
- Bilingual
- Courtesy Patrol
- Trash Pickup - Curbside
- Recycling
- Planned Social Activities
- Pet Play Area

Kitchen

- Dishwasher
- Disposal
- Ice Maker
- Granite Countertops
- Kitchen
- Microwave
- Oven
- Refrigerator
- Freezer
- Instant Hot Water

Property Information

- Built in 2015
- 442 Units/2 Stories

Lease Length

August 1 - July 22

Outdoor Space

- Grill
- Waterfront
- Pond

Features

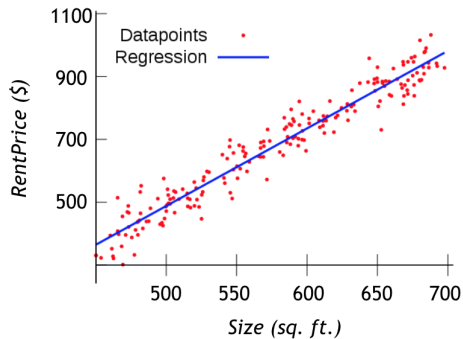
- High Speed Internet Access
- Washer/Dryer
- Air Conditioning
- Heating
- Ceiling Fans
- Smoke Free
- Cable Ready
- Tub/Shower
- Framed Mirrors

Source: apartments.com

$$RentPrice = w_0 + w_1 \times Size + w_2 \times DistanceFromCS + \dots$$

Linear Regression: Example

$$\text{RentPrice} = w_0 + w_1 \times \text{Size} + w_2 \times \text{DistanceFromCS} + \dots$$



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$$\text{RentPrice} = w_0 + w_1 \times \text{Size} + w_2 \times \text{DistanceFromCS} + \dots$$

How do we find the unknown model parameters $\{w_0, w_1, w_2, \dots\}$?

We use training data!

Training Sample	Size (sq.ft.)	DistanceFromCS (miles)	RentPrice (\$)
1	498	11.9	675
2	513	8.6	750
3	621	8.3	800
4	710	3.4	965
...	...	...	...